AWG19 POLARIZATION DRIVER — CONNECTED SCHEMATIC

External coil: 11.430016 ohm / 25.92669 mH
Polarize: Q1+Q4 on. Q5 off.
Discharge: Q1+Q4 off, clamp conducts.
Receive: Q5 on, U2A on, other U2 switches off.

Q4 / Q5 need MCU dead time.
No hardware interlock yet.
Regulated 12 V supply required.

LOCAL BYPASS: place each capacitor at its named IC supply pin

Sensor: 1.65 V at zero current; 1 V/A.
3.3 V MCU ADC assumed.

J5 raw differential: 50 mV/A.
J3 I_ADC remains amplified.
Polarization current only.

DRAFT — component ratings / footprints still need review.
Current monitor measures polarization, not clamp current.
Kelvin route U4 input pair to R9 pads.
Signal and MCU labels join only the named remote nets.

U3 LTC1693-5
U1 LTC1693-1
J1 REGULATED SUPPLIES
J2 EXTERNAL AWG19
J3 MCU CONTROL / ADC
J4 TO AMPLIFIER
J5 RAW SHUNT SENSE
U2A U2B U2C U2D U2E ADG5412F
U4 INA240A1PW
Q1 Q4 Q5 IRF530 / IRF9640
R1 R2 R3 R4 R5 R6 R7 R8 R9 R10
C1 C2 C3 C4 C5 C6 C7 C8
CLAMP_MID
FAST — TBD2 TVS 43V — TBD1

ECE 455 — Jun Yang	
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